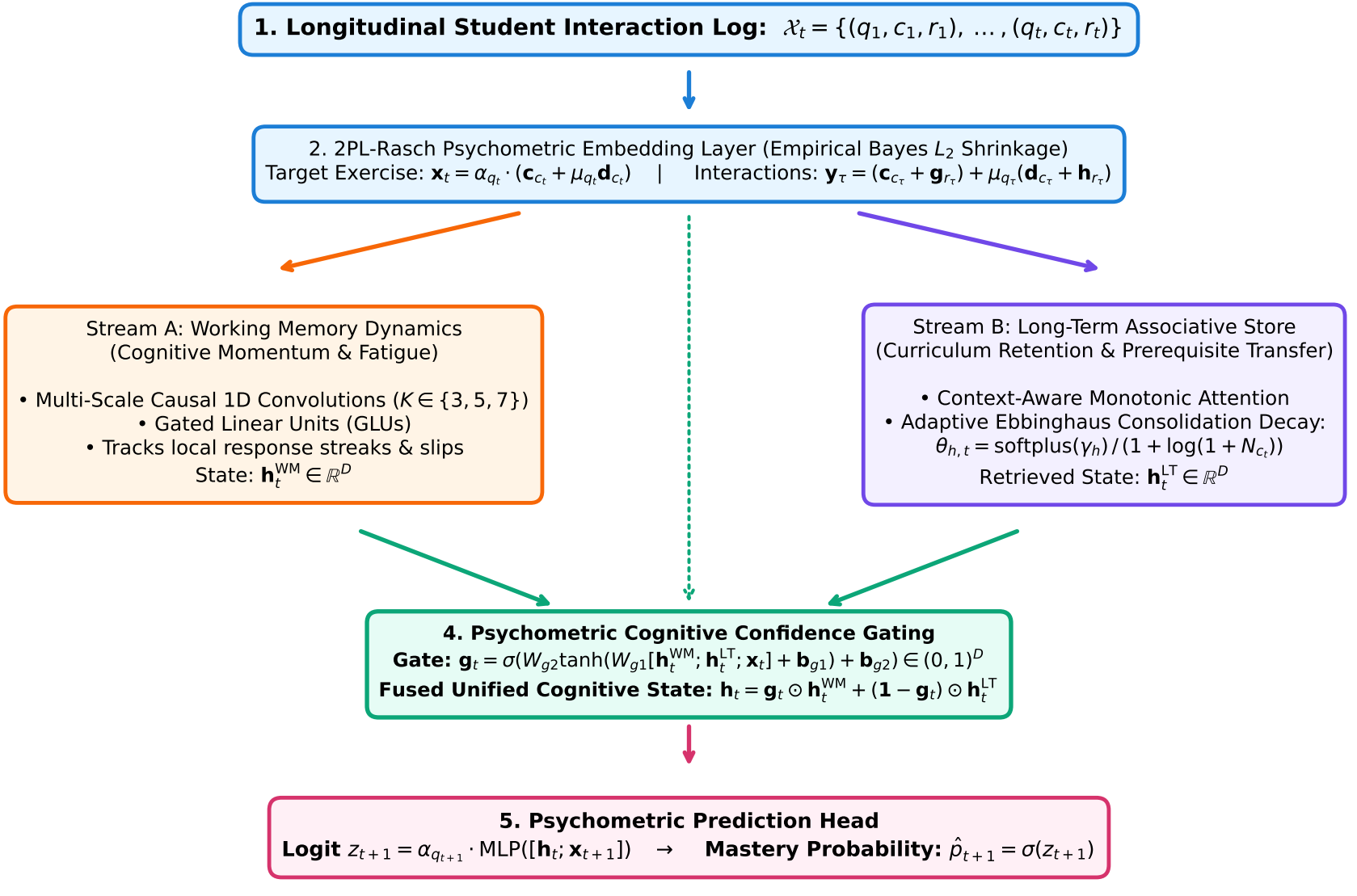
1. Longitudinal Student Interaction Log: $\mathcal{X}_t = \{(q_1, c_1, r_1), \ldots, (q_t, c_t, r_t)\}$
2. 2PL-Rasch Psychometric Embedding Layer (Empirical Bayes L_2 Shrinkage)
Target Exercise: $\mathbf{x}_t = \alpha_{q_t} \cdot (\mathbf{c}_{c_t} + \mu_{q_t}\mathbf{d}_{c_t})$ | Interactions: $\mathbf{y}_\tau = (\mathbf{c}_{c_\tau} + \mathbf{g}_{r_\tau}) + \mu_{q_\tau}(\mathbf{d}_{c_\tau} + \mathbf{h}_{r_\tau})$
Stream A: Working Memory Dynamics
(Cognitive Momentum & Fatigue)
• Multi-Scale Causal 1D Convolutions ($K \in \{3, 5, 7\}$)
• Gated Linear Units (GLUs)
• Tracks local response streaks & slips
State: $\mathbf{h}_t^{\text{WM}} \in \mathbb{R}^D$
Stream B: Long-Term Associative Store
(Curriculum Retention & Prerequisite Transfer)
• Context-Aware Monotonic Attention
• Adaptive Ebbinghaus Consolidation Decay:
$\theta_{h,t} = \text{softplus}(\gamma_h) / (1 + \log(1 + N_{c_t}))$
Retrieved State: $\mathbf{h}_t^{\text{LT}} \in \mathbb{R}^D$
4. Psychometric Cognitive Confidence Gating
Gate: $\mathbf{g}_t = \sigma(W_{g2}\tanh(W_{g1}[\mathbf{h}_t^{\text{WM}}; \mathbf{h}_t^{\text{LT}}; \mathbf{x}_t] + \mathbf{b}_{g1}) + \mathbf{b}_{g2}) \in (0, 1)^D$
Fused Unified Cognitive State: $\mathbf{h}_t = \mathbf{g}_t \odot \mathbf{h}_t^{\text{WM}} + (\mathbf{1} - \mathbf{g}_t) \odot \mathbf{h}_t^{\text{LT}}$
5. Psychometric Prediction Head
Logit $z_{t+1} = \alpha_{q_{t+1}} \cdot \text{MLP}([\mathbf{h}_t; \mathbf{x}_{t+1}])$ → Mastery Probability: $\hat{p}_{t+1} = \sigma(z_{t+1})$